

Decoding Kryptos K2 the Analog Visual Way

A paper and pencil method to decrypt the Kryptos K2 Vigenère style cipher

August 28, 2019

Examination of the ciphertext for patterns

The entire K2 is first laid out and numbered by position and then analyzed for repeating segments.

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K
21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
K	Q	E	T	G	F	Q	J	N	C	E	G	G	W	H	K	K	D	Q	M
41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
C	P	F	Q	Z	D	Q	M	M	I	A	G	P	F	X	H	Q	R	L	G
61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80
T	I	M	V	M	Z	J	A	N	Q	L	V	K	Q	E	D	A	G	D	V
81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
F	R	P	J	U	N	G	E	U	N	A	Q	Z	G	Z	L	E	C	G	Y
101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
U	X	U	E	E	N	J	T	B	J	L	B	Q	C	R	T	B	J	D	F
121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140
H	R	R	Y	I	Z	E	T	K	Z	E	M	V	D	U	F	K	S	J	H
141	142	143	144	145	146	147	148	149	150	151	152	153	154	155	156	157	158	159	160
K	F	W	H	K	U	W	Q	L	S	Z	F	T	I	H	H	D	D	D	U
161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
V	H	D	W	K	B	F	U	F	P	W	N	T	D	F	I	Y	C	U	Q
181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200
Z	E	R	E	E	V	L	D	K	F	E	Z	M	O	Q	Q	J	L	T	T
201	202	203	204	205	206	207	208	209	210	211	212	213	214	215	216	217	218	219	220
U	G	S	Y	Q	P	F	E	U	N	L	A	V	I	D	X	F	L	G	G
221	222	223	224	225	226	227	228	229	230	231	232	233	234	235	236	237	238	239	240
T	E	Z	F	K	Z	B	S	F	D	Q	V	G	O	G	I	P	U	F	X
241	242	243	244	245	246	247	248	249	250	251	252	253	254	255	256	257	258	259	260
H	H	D	R	K	F	F	H	Q	N	T	G	P	U	A	E	C	N	U	V
261	262	263	264	265	266	267	268	269	270	271	272	273	274	275	276	277	278	279	280
P	D	J	M	Q	C	L	Q	U	M	U	N	E	D	F	Q	E	L	Z	Z
281	282	283	284	285	286	287	288	289	290	291	292	293	294	295	296	297	298	299	300
V	R	R	G	K	F	F	V	O	E	E	X	B	D	M	V	P	N	F	Q
301	302	303	304	305	306	307	308	309	310	311	312	313	314	315	316	317	318	319	320
X	E	Z	L	G	R	E	D	N	Q	F	M	P	N	Z	G	L	F	L	P
321	322	323	324	325	326	327	328	329	330	331	332	333	334	335	336	337	338	339	340
M	R	J	Q	Y	A	L	M	G	N	U	V	P	D	X	V	K	P	D	Q
341	342	343	344	345	346	347	348	349	350	351	352	353	354	355	356	357	358	359	360
U	M	E	B	E	D	M	H	D	A	F	M	J	G	Z	N	U	P	L	G
361	362	363	364	365	366	367	368	369											
E	W	J	L	L	A	E	T	G											

The summary of the fragments are as follows:

Repeating section	1 st location	2 nd location	Difference
PJU	3	83	80
GWHK	17	33	16
DQM	38	46	8
TBJ	108	116	8
EZ	222	302	120
KFF	245	285	40
QUM	268	340	72

All the repeating sections appear to be at a distance as a multiple of 8. Repeating segments of DQM and TBJ are the smallest distances apart (8 characters) and very likely this would be the periodicity of the cipher. Moreover, the countless number of repeating segments would support some form of a substitution cipher rather than a transposition because the position of the plaintext and cipher text appears to be maintained and the number of repeating segments identified.

Stenographic Clues in Kryptos models

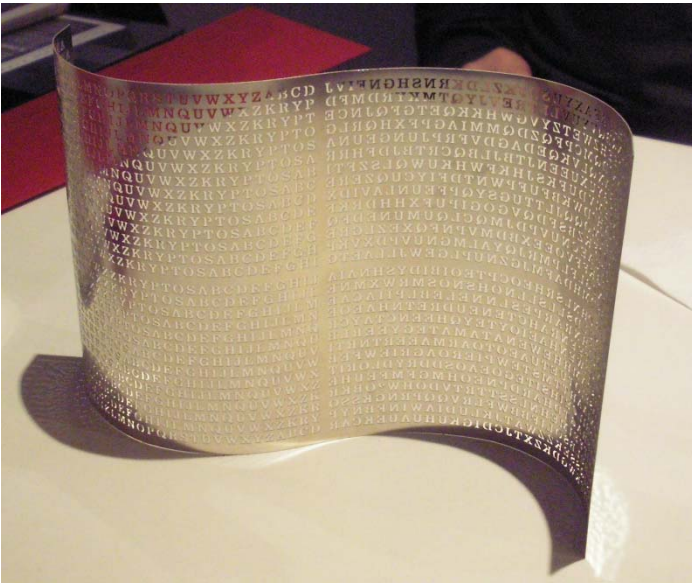
K1 utilized the top half of the KRYPTOS tableau to provide the plaintext order (i.e. KRYPTOSABC...) for the Quagmire III decode. Using the same pattern of stenography, it is very difficult to conclude exactly which portions of the tableau is used on the CIA sculpture. That being said, if Jim Sanborn intended on the use of stenography on the tableau, then the information would be preserved in other smaller scaled Kryptos replicas.

Mini-Kryptos, 2005 Jim Sanborn dinner models (larger) and the studio tabletop Kryptos all indicate the L and Z portion of the bottom tableau to be of significance (see below).

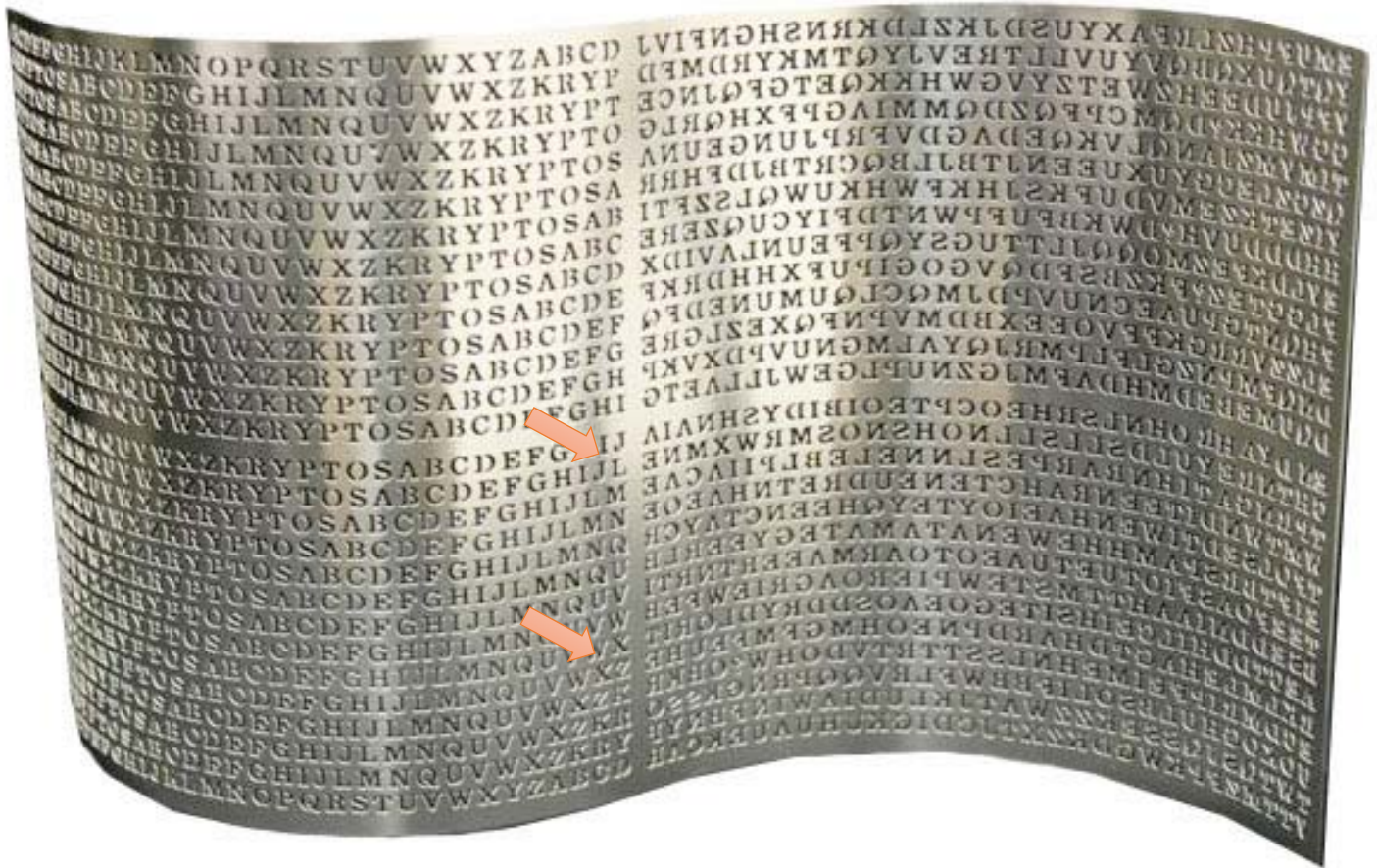
From L to Z on the tableau, it is 9 lines or a 1+8 format. That is, the tableau is the 9 lines with the first being the plain alphabet character ordering and the subsequent 8 being the actual cipher alphabet (by rows).



Kryptos model from Jim Sanborn's studio (Great Big Story 2019)



Above: Larger Kryptos model from 2005 dinner (Chris "Xenon")



Mini-Kryptos model – L & Z at the right margin of the bottom half of tableau are indented.



Jim Gillogly's tableau image with the "L" inside the right margin and "W" extended past the left margin

	THE CODE	THE KEY
K1	EMUFPHZLRFAXYUSDJKZLDRNSHGNFIVJ YQTQUXQBQVYUULLTREVJYQTMKYRDMFD VFPJUDEEHZWETZYVGWHKKQETGFQJNCE GGWHKK?DQMCPPQZDQMMIAGPFHXQRLG TIMVMZJANQLVKQEDAGDVFRPJUNGEUNA QZGZLECGYUXUEENJTBJLBQCRTBJDFHRR YIZETKZEMVDFKSHKFWHQUWQLSZFTI HHDDDUVH?DWKBFUFPWNTDFIYCUQZERE EVLDKFEZMOQQJLTUGSYQPFUNLAVIDX FLGGTEZ?FKZBSFDQVGOGIPUFXHHRKRF FHQNTGPAECNUVPDQJMQCLQUMUNEDFQ ELZZVRRGKFFVOEEBXDMVPNFQXEZLGRE DNQFMPNZGLFLPMRJQYALMGNUVPDXVKP DQUMEBEDMHDAFMJGZNUPLGEWJLLAETG	ABCDEFGHIJKLMNOPQRSTUVWXYZABCD AKRYPTOSABCDEFGHIJKLMNQVWXYZKRYPT BRYPTOSABCDEFGHIJKLMNQVWXYZKRYPT CYPTOSABCDFFGHIJKLMNQVWXYZKRYPTO DPTOSABCDEFGHIJKLMNQVWXYZKRYPTOS ETOSABCDEFGHIJKLMNQVWXYZKRYPTOSA FOSABCDEFGHIJKLMNQVWXYZKRYPTOSAB GSABCDEFGHIJKLMNQVWXYZKRYPTOSABC HABCDEFGHIJKLMNQVWXYZKRYPTOSABCD IBCDEFGHIJKLMNQVWXYZKRYPTOSABCDE JCDEFGHIJKLMNQVWXYZKRYPTOSABCDEF KDEFGHIJKLMNQVWXYZKRYPTOSABCDEF LEFGHIJKLMNQVWXYZKRYPTOSABCDEF MFGHIJKLMNQVWXYZKRYPTOSABCDEF
K2	ENDY AHR OHNLSRHEOCPTIOIBIDYSHNAIA CHTNREYULDSLLSLLNOHSNOSMRWXMNE TPRNGATIHNRRARPESLNNELEBLPIACAE WMTWNDITEENRAHCTENEUDRETNAEOE TFOLSEDTIWENHAEIOYTEYQHEENCTAYCR EIFTBRSAPAMHHEWENATAMATEGYEERLB TEEFOASFOTUETUAEOARMAEERTNRTI BSEDDNIAAHTTMSTEWPIEROAGRIEVEB AECTDDHILCEIHSITEGOEAOSSDDRYDLO RKLMLHAGTDHARDPNEOHMGFMFEUHE ECDMRIPFEIMEHNLSSTTRTVDOHW?OBKR UOXOGHULBSOLIFBBWFLRVQQPRNGKSSO TWTQSJSSEKZZWATJKLUDIAWINFBNYP VTTMZFPKWGDKZXTJCDIGKUHUAUEKCAR	NGHIJLMNQVWXYZKRYPTOSABCDEFGHIJL OHIJLMNQVWXYZKRYPTOSABCDEFGHIJL PIJLMNQVWXYZKRYPTOSABCDEFGHIJLM QJLMNQVWXYZKRYPTOSABCDEFGHIJLMN RLMNQVWXYZKRYPTOSABCDEFGHIJLMNQ SMNQVWXYZKRYPTOSABCDEFGHIJLMNQ TNQVWXYZKRYPTOSABCDEFGHIJLMNQV UQVWXYZKRYPTOSABCDEFGHIJLMNQVW VUVWXYZKRYPTOSABCDEFGHIJLMNQVW WVWXYZKRYPTOSABCDEFGHIJLMNQVW XWYZKRYPTOSABCDEFGHIJLMNQVW YXZKRYPTOSABCDEFGHIJLMNQVW ZKRYPTOSABCDEFGHIJLMNQVW ABCDEFGHIJKLMNOPQRSTUVWXYZABCD
K3		
K4		

Wired (2005) image for Kryptos with the L and W stenographic clue

Similar to K1 stenographic clues that utilize the P & G on the right margin to indicate the deciphering tableau, the K2 tableau can be found by spotting the changes in the margin. The change in margin alignment from the CIA Kryptos to the smaller scaled models was likely due to making the stenographic clue much more apparent.

Building the encoding/decoding tableau

From the L and 4 characters back, the decoding tableau would be shown as:

	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G
1	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z
2	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z
3	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z
4	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z
5	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z
6	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z
7	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z
8	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z

For rows 1 through 8 there would be a Caesar shift by “x” units. While it may seem difficult to determine the plaintext from the tableau manually, the symmetrical nature of classical substitution is the secret to encoding and decoding. That is, the encoding and decoding is done by the exact same process of utilizing the tableau. If certain relationships can be identified, then the plaintext can be predicted to produce the decoded message.

Similar to the K1 Quagmire III decode, the most important weakness of a Vigenère style tableau is that the polyalphabetic substitution is on a 1:1 basis. For each row, K plaintext can only be encoded to a single character (A-Z) and the character shifts between the plaintext or ciphertext rows must be maintained. As an example, the number of shifts from K to B is 8 (K to RYPTOSAB). This is identical on the ciphertext rows regardless of the offset.

For the purposes of the manual method, the order of HIJL...EFG of the plaintext ordering will be maintained to match the process of K1. This might be useful to provide some clue for K4 in the future. The next step is to mark down all the letters used by the ciphertext:

	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G
1	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z
2	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z
3	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z
4	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z
5	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z
6	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z
7	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z
8	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z

The banding patterns are difficult to distinguish because the weighting of the ciphertext letters is not factored in above. To deal with this, the ciphertext letter counts will have to be tabulated.

Frequency counting of ciphertext letters

If some patterns in the plaintext/ciphertext can be determined, then a partial crib might be possible. Generally, English language has minimal use of the letters “X”, “Q” and “Z”, and significant usage of the vowels and some consonants such as the letters S and T. By identifying the highest frequency of ciphertext letters, it may correspond to commonly used English letters (AEIOURSTLN) on the plaintext order.

From above, it was determined the periodicity of the substitution cipher was 8 characters by the repeating segments and stenographic clues. If the ciphertext letters are tallied up:

Letter	Row1	Row2	Row3	Row4	Row5	Row6	Row7	Row8	Grand Total
A			2	2	1	3	1		9
B			1		3	1		2	7
C	2	4				1			7
D		2	3	3	2	10	4		24
E	6	1	4	1	1	3	7	6	29
F	3	2	3		1	6	8	4	27
G	7	2	3	6		2	1	2	23
H	3	2	3	3				3	14
I		2		1	1	2		1	7
J	1		5	3	1	2	1		13
K	6			2	7				15
L		1	4	1	3	2	5	2	18
M	4		1	1		2	2	6	16
N		8		2	3	1		1	15
O	1	2							3
P	2	3	2		5	2		1	15
Q	4	1	2	10	1	3	3		24
R		6	3	1			1		11
S		1	1	1		1			4
T	1		1	2	4		1	4	13
U	3	1	3		6	2	3	2	20
V	3	1		2	2			8	16
W		3	3	1			1		8
X				1	1	1	2	2	7
Y	1			3	1		1		6
Z		4	2		3	2	5	2	18
Grand Total	47	46	46	46	46	46	46	46	369
Segment	KEG	RCNZ	EJL	YDGJQ	KPU	ADEFQ	EFLZ	EMV	

At least three characters are identified for each “row” of the tableau. When the third highest letter for frequency is a tie, then all letters are included. This occurs for rows 4 (DJY) and 6 (AEQ).

The objective is to create banding patterns that would align with plaintext letters of high usage to possibly crib decode K2.

The high frequency count of ciphertext letters will likely correspond to high plaintext letters. Vowels AEIOU and consonants RSTLN might be used more frequently. Inspiration of the use RSTLN comes from the USA game show *Wheel of Fortune* “Bonus Round” where the constant will pick a word puzzle and have RSTLNE characters given to them.

[illegible]

	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G
1	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z
2	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z
3	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z
4	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z
5	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z
6	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z
7	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z
8	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z

	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G
1	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q
2	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U
3	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W
4	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V
5	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F
6	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N
7	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N
8	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q

Page 7 of 10

Analysis of partially decoded plaintext and prediction of letters

1	2	3	4	5	6	7	8
I	T	N	A	V	T	O	T
A	L	G	Y	O	N	V	I
S	I	T	L	R	H	O	W
S	T	D	A	Q	P	O	S
S	I	T	L	R	T	H	E
Y	U	Y	E	K	T	H	E
E	A	W	T	T	S	M	A
G	N	A	T	O	C	F	I
E	L	S	X	Q	H	E	I
N	F	R	R	B	A	T	I
O	N	N	A	V	G	A	T
H	E	W	E	K	A	N	D
T	R	P	N	V	M	I	T
T	E	S	U	C	D	E	R
G	R	L	U	C	D	T	O
A	N	L	N	J	N	O	W
N	L	R	C	W	T	I	O
N	X	S	O	R	S	L	A
N	G	G	E	M	K	N	O
W	A	T	O	E	T	T	H
I	S	K	H	R	Y	S	H
O	U	G	D	O	T	S	B
U	R	E	E	K	O	U	T
T	H	A	R	R	S	O	M
E	W	D	E	L	E	X	W
H	O	V	N	U	W	S	T
H	E	A	X	W	C	T	L
O	C	P	T	O	O	N	O
N	L	X	W	G	T	H	I
S	W	P	S	T	I	S	L
A	S	K	M	R	S	S	A
G	E	Q	T	T	I	R	T
Y	E	E	G	T	T	D	E
G	R	A	E	V	F	I	F
T	Y	Y	E	F	E	N	M
I	N	L	T	R	S	S	I
X	P	R	I	C	T	F	I
V	E	Y	E	Z	O	N	D
S	N	R	R	Q	H	S	E
V	E	I	T	M	S	E	V
E	N	S	E	P	R	E	E
S	E	E	G	T	T	M	I
N	U	K	E	V	F	O	R
T	Y	B	O	E	R	S	E
C	O	I	D	V	W	E	S
T	I	S	B	M	R	O	W
S							

Rows 3 & 5 do not appear to be correctly aligned for the decryption and will be removed

Yellow highlighted text indicates possible complete words

Green highlighted text indicates possible words that can be completed through educated guessing

1	2	3	4	5	6	7	8
I	T		A		T	O	T
A	L		Y		N	V	I
S	I		L		H	O	W
S	T		A		P	O	S
S	I		L		T	H	E
Y	U		E		T	H	E
E	A		T		S	M	A
G	N		T		C	F	I
E	L		X		H	E	I
N	F		R		A	T	I
O	N		A		G	A	T
H	E		E		A	N	D
T	R		N		M	I	T
T	E		U		D	E	R
G	R		U		D	T	O
A	N		N		N	O	W
N	L		C		T	I	O
N	X		O		S	L	A
N	G		E		K	N	O
W	A		O		T	T	H
I	S		H		Y	S	H
O	U		D		T	S	B
U	R		E		O	U	T
T	H		R		S	O	M
E	W		E		E	X	W
H	O		N		W	S	T
H	E		X		C	T	L
O	C		T		O	N	O
N	L		W		T	H	I
S	W		S		I	S	L
A	S		M		S	S	A
G	E		T		I	R	T
Y	E		G		T	D	E
G	R		E		F	I	F
T	Y		E		E	N	M
I	N		T		S	S	I
X	P		I		T	F	I
V	E	S	E	C	O	N	D
S	N		R		H	S	E
V	E		T		S	E	V
E	N		E		R	E	E
S	E		G		T	M	I
N	U	T	E	S	F	O	R
T	Y		O		R	S	E
C	O	N	D	S	W	E	S
T	I		B		R	O	W
S							

By examining the bottom section of the plaintext, there appears to be plaintext describing numbers and time a possible crib might be possible by shifting row 3 and 5 to match the guessed letters

It is important to note that there are repeated ciphertext letters so the row 5 "S" decrypt must be the same

With the predicted number and time plaintext letters, this would imply that for Row 3 the ciphertext letters of FDZ should correspond to STN and Row 5 XUJ should correspond to CSS. Row 3 and 5 is adjusted (shifted) to:

	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G
1	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q
2	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U
3	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N
4	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V
5	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y
6	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N
7	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N
8	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q

The resulting decode yields the following:

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
1	I	T	W	A	S	T	O	T	A	L	L	Y	I	N	V	I	S	I	B	L	E	H	O	W
2	S	T	H	A	T	P	O	S	S	I	B	L	E	T	H	E	Y	U	S	E	D	T	H	E
3	E	A	R	T	H	S	M	A	G	N	E	T	I	C	F	I	E	L	D	X	T	H	E	I
4	N	F	O	R	M	A	T	I	O	N	W	A	S	G	A	T	H	E	R	E	D	A	N	D
5	T	R	A	N	S	M	I	T	T	E	D	U	N	D	E	R	G	R	U	U	N	D	T	O
6	A	N	U	N	K	N	O	W	N	L	O	C	A	T	I	O	N	X	D	O	E	S	L	A
7	N	G	L	E	Y	K	N	O	W	A	B	O	U	T	T	H	I	S	T	H	E	Y	S	H
8	O	U	L	D	I	T	S	B	U	R	I	E	D	O	U	T	T	H	E	R	E	S	O	M
9	E	W	H	E	R	E	X	W	H	O	K	N	O	W	S	T	H	E	E	X	A	C	T	L
10	O	C	A	T	I	O	N	O	N	L	Y	W	W	T	H	I	S	W	A	S	H	I	S	L
11	A	S	T	M	E	S	S	A	G	E	X	T	H	I	R	T	Y	E	I	G	H	T	D	E
12	G	R	E	E	S	F	I	F	T	Y	S	E	V	E	N	M	I	N	U	T	E	S	S	I
13	X	P	O	I	N	T	F	I	V	E	S	E	C	O	N	D	S	N	O	R	T	H	S	E
14	V	E	N	T	Y	S	E	V	E	N	D	E	G	R	E	E	S	E	I	G	H	T	M	I
15	N	U	T	E	S	F	O	R	T	Y	F	O	U	R	S	E	C	O	N	D	S	W	E	S
16	T	I	D	B	Y	R	O	W	S															

By inspecting the manipulated rows vertically, the only lettering available that would spell an actual word with all 8 characters is ABSCISSA under the plaintext alphabet "K". The orientation of the tableau is reminiscent of Quagmire III.

Discussion and Analysis – Comparison to K1

Plaintext Alphabet Alignment

Similar to K1, K2 was decoded from a substitution Vigenere style cipher that exploited the symmetrical nature of the encryption and decryption. Both ciphertext and plaintext utilize the identical key of ABSCISSA via Quagmire III cipher. To match the original tableau, it may be of importance to maintain the alignment of the plaintext alphabet where KRYPTOS is in the middle of the tableau rather than at the beginning like Quagmire III.

IDBYROWS vs XLAYERTWO

The final 8 characters yields IDBYROWS which may or may not have been intentional due to Jim Sanborn's K2 correction. Following Jim's "onion-layer" theory, it is possible that Jim intentionally omitted the X for the decryption as both a clue and a checkpoint. In terms of a clue, the IDBYROWS might be prompt for the K3 transposition while Jim may have intentionally withheld the XLAYERTWO until he felt the Kryptos community reached a certain stage in the decryption.

Kryptos Tableau

By identifying the tableau sections for K1 and K2, there are two sections in the tableau that have yet to be used. It may be possible that there is information to deal with K3 and K4 but may have to be explored in the future.

	THE CODE	THE KEY
K1	EMUFPHZLRFAXYUSDJKZLDKRN SHGNFIVJ YQTQUXQBQVYUULLTREVVJYQTMKYRDMFD VFPJUDEEHZWETZYVGWHKKQETGFGJNCE GGWHKK?DQMCPPFQZDQMMIAGPFXHQRLG TIMVMZJANQLVKQEDAGDVFRPJUNGEUNA QQZGLECGYUXUEENJTBBLBQCRTBJDFHRR YIZETKZEMVDUFKSJHKFWHKUWQLSZFTI HHDDDUVH?DWKBFUFPWNTDFIYCUQZERE EVLDFEFZMOQQJLTTUGSYQPFUNLAVIDX FLGGTEZ?FKZBSFDQVGOGIPUFXXHHRKF FHQNTGPPUAECNUVPDJMQCLQUMUNEDFQ ELZZVRRGKFFVOEEXBDMVFNFXQEZLGRE DNQFMPNZGLFLPMRJQYALMGNUVPDXVKP DQUMEBEDMDHDAFMJGZNUPLGEWJLLAETG	ABCDEFGHIJKLMN O PQRSTU VWXYZ ABCD AKRYPTOSABCDEFGHIJLMNQUVWXXZKRYPT BRYPTOSABCDEFGHIJLMNQUVWXXZKRYPT CYPTOSABCDEFGHIJLMNQUVWXXZKRYPT DPTOSABCDEFGHIJLMNQUVWXXZKRYPT ETOSABCDEFGHIJLMNQUVWXXZKRYPT FOSABCDEFGHIJLMNQUVWXXZKRYPT GSABCDEFGHIJLMNQUVWXXZKRYPT HABCDEFGHIJLMNQUVWXXZKRYPT IABCDEFGHIJLMNQUVWXXZKRYPT JABCDEFGHIJLMNQUVWXXZKRYPT KABCDEFGHIJLMNQUVWXXZKRYPT LABCDEFGHIJLMNQUVWXXZKRYPT MABCDEFGHIJLMNQUVWXXZKRYPT
K2	ENDY AHR OHNLSRHEOCPTEOIBIDYSHNAIA CHTNREYULDSL LSLN OHSNOSMRWXMNE TPRNGATIHNRARPESLNNELEBLPIIACAE WMTWNDITEENRAHCTENEUDRETNAEOE TFOLSEDTIWENHAEIOYTEYQHEENCTAYCR EIFTBRS PAMHHEWENATAMATEGYEEERLB TEEFOASFOTUETUAEO TOARMAEEERTNRTI BSEDDNIAAHTTMSTEWP IEROAGRIEWFEB AECTDDHILCEIHSITEGOEAO SDDRYDLORIT RKLMLHEAGTDHARDPNEOHMGFMFEUHE ECDMRIPFEIMEHNLSSTTRTVDOHW?OBKR UOXGHULBSOLIFBBWFLRVQQPRNGKSSO TWTQSQJSSEKZZWATJKLUDIAWINFBNYP VTTMZFPK WGDKZXTJCDIGKUHUAUEKCAR	NGHIJLMNQUVWXXZKRYPTOSABCDEFGHIJL OHIJLMNQUVWXXZKRYPTOSABCDEFGHIJL PIJLMNQUVWXXZKRYPTOSABCDEFGHIJLM QJLMNQUVWXXZKRYPTOSABCDEFGHIJLMN RLMNQUVWXXZKRYPTOSABCDEFGHIJLMNQ SMNQUVWXXZKRYPTOSABCDEFGHIJLMNQ TNQUVWXXZKRYPTOSABCDEFGHIJLMNQV UQUVWXXZKRYPTOSABCDEFGHIJLMNQVW VUVWXXZKRYPTOSABCDEFGHIJLMNQVWX WVWXZKRYPTOSABCDEFGHIJLMNQVWXZ XWXZKRYPTOSABCDEFGHIJLMNQVWXZK YXZKRYPTOSABCDEFGHIJLMNQVWXZKR ZZKRYPTOSABCDEFGHIJLMNQVWXZKRY ABCDEFGHIJKLMN O PQRSTU VWXYZ ABCD
K3		
K4		

K4 Predictions

Similar to the K1 manual decrypt for substitutions, K4 could be much more difficult if the substitution attacks are mitigated by a non-cyclical asymmetrical key without the use of plain English. Combined with a transposition cipher, it would be extremely difficult to decrypt. As mentioned in the C2 K4-Double-Mask-OTP hypothesis (July 15, 2019), a possible decryption process would be the transposition of K4 characters and then the decryption via one-time pad (Vernam, XOR, etc.). An unorthodox transposition would be best for Jim Sanborn to leverage his creativity.